

Adaptive Game Model - Technical Report

Event-based tactical decision intelligence: methodology, validation, and reproducibility

1. Problem & modelling decision

We estimate which build-up adjustment should be used against a specific opponent structure - not which team wins. Every possession is labelled with a 0-5 outcome scale and converted to expected possession value (EPV) in goal-probability units. The value model (XGBoost + isotonic calibration) is trained on ex-ante features; tactical states are discovered by clustering and named by football prototypes; adjustment values are estimated causally with stabilised IPW and a doubly-robust (AIPW) augmentation, with match-clustered bootstrap confidence intervals.

2. Data & preprocessing

Source: statsbomb_open_data | 40 matches, 7582 possessions.

Check	Severity	Count	Message
duplicate_events	info	0	No duplicate events
impossible_coordinates	info	0	All coordinates within pitch
missing_player_ids	info	0	0/87756 on-ball events lack player_id (0.0%)
invalid_timestamps	info	0	All timestamps valid
team_name_consistency	info	0	Team names consistent with manifest
competition_mismatch	info	0	All events map to manifest matches
outliers	warning	34	34 passes longer than 90m
leakage	info	0	No match-level leakage between train and test

3. Possession value model

Model	RMSE	MAE	R2
xgboost_calibrated	0.0521	0.0318	0.143
xgboost_raw	0.0528	0.0323	0.121
linear	0.0594	0.0330	-0.114
elo_only	0.0564	0.0342	-0.005
league_average	0.0564	0.0342	-0.005

4. Calibration & temporal validation

Out-of-time calibration ECE = 0.0040 (MSE 0.00272). The model is evaluated strictly chronologically: train on earlier matches, validate and test on later matches. No action from a match appears in more than one split (leakage check passes).

5. Tactical states

Opponent high press - 3581 sequences, baseline EPV 0.010.

Final-third attack vs compact defence - 1292 sequences, baseline EPV 0.059.

Wide overload - 1100 sequences, baseline EPV 0.087.

Opponent low block - 1091 sequences, baseline EPV 0.024.

Neutral build-up / rest state - 518 sequences, baseline EPV 0.007.

6. Adjustment value estimates

State	Adjustment	Uplift	CI	Confidence
Final-third attack vs compact defence	s_third_man_combination	+0.012	[+0.005, +0.018]	high
Final-third attack vs compact defence	s_aggressive_counterpress	-0.005	[-0.010, +0.001]	medium
Neutral build-up / rest state	s_mid_block_retreat	+0.016	[+0.014, +0.020]	medium
Opponent high press	s_aggressive_counterpress	-0.000	[-0.003, +0.002]	medium
Opponent high press	s_mid_block_retreat	-0.001	[-0.004, +0.001]	medium
Opponent high press	s_third_man_combination	-0.005	[-0.008, -0.002]	medium
Opponent low block	s_aggressive_counterpress	+0.002	[-0.005, +0.009]	medium
Opponent low block	s_third_man_combination	-0.012	[-0.030, +0.013]	medium
Wide overload	s_aggressive_counterpress	+0.008	[-0.016, +0.032]	medium
Wide overload	s_third_man_combination	-0.001	[-0.027, +0.023]	medium

7. Error analysis

Three worst failures:

Barcelona vs None (min 89): actual 0.603 vs predicted 0.037 - Model expected a routine possession but a 0.60-xG chance arrived from a unknown scenario - high-variance outcomes from sparse rare events.

Atlético Madrid vs None (min 6): actual 0.575 vs predicted 0.058 - Model expected a routine possession but a 0.57-xG chance arrived from a unknown scenario - high-variance outcomes from sparse rare events.

Atlético Madrid vs None (min 47): actual 0.452 vs predicted 0.037 - Model expected a routine possession but a 0.45-xG chance arrived from a unknown scenario - high-variance outcomes from sparse rare events.

8. Ablation & sensitivity

Feature group	RMSE	MAE	R2
full	0.0521	0.0319	0.144
no_elo	0.0522	0.0319	0.141
no_match_context	0.0521	0.0320	0.143
no_location	0.0528	0.0322	0.121

9. Known limitations

- Event-based: no player tracking data is used.
- Adjustment labels are transparent rule-based proxies of coaching concepts.
- Estimates are historical expectations with uncertainty, not guarantees.
- Applicability limited to opponents/leagues represented in training data.
- Small samples yield low-confidence recommendations by design.

10. Reproducibility

Run **python -m src.pipeline.run_all** to reproduce all artefacts. Data manifest with SHA-256 fingerprints lives in `data/processed/data_manifest.json`. Random seed fixed (42); synthetic fallback is deterministic for offline CI.